

ABM BUSINESS LOGIC BIBLE

SIGNAL SOURCE REGISTRY — EXHAUSTIVE SPECIFICATION

Every source named | Authenticated | Timed | Per-account specific | EPIS-profiled

1. TARGET ACCOUNT UNIVERSE

The engine monitors a defined set of KSA banking and financial institutions. Every source below is configured per-account — not as a generic scan. The following are the primary targets referenced across the spec:

Account	Tadawul Ticker	Domain	LinkedIn Co. Page	IT Subsidiary	Careers URL
Saudi National Bank (SNB)	1180	alahli.com	linkedin.com/company/snbalahli	ITQAN	alahli.com/careers
Al Rajhi Bank	1120	alrajhibank.com.sa	linkedin.com/company/alrajhi-bank	EJADA	alrajhibank.com.sa/en/careers
Riyad Bank	1010	riyadbank.com	linkedin.com/company/riyad-bank	—	riyadbank.com/en/careers
Saudi British Bank (SAB)	1060	sab.com	linkedin.com/company/saborig	—	sab.com/en/careers
Alinma Bank	1150	alinma.com	linkedin.com/company/alinma-bank	—	alinma.com/en/careers
Arab National Bank (ANB)	1080	anb.com.sa	linkedin.com/company/arab-national-bank	—	anb.com.sa/en/careers
Bank AlJazira (BAJ)	1020	baj.com.sa	linkedin.com/company/bank-aljazira	—	baj.com.sa/careers
Bank Albilad	1140	bankalbilad.com	linkedin.com/company/bank-albilad	—	bankalbilad.com/en/careers
STC Pay / stc bank	—	stcbank.com.sa	linkedin.com/company/stc-pay	—	stcbank.com.sa/careers
Saudi Digital Bank (D360)	—	d360bank.com	linkedin.com/company/d360bank	—	d360bank.com/careers

Neo (SNB subsidiary), Emkan Finance, Tiqmo, and other digital banking licensees are tracked as subsidiary entities under their parent account (SIG-SUBS). Fintech ecosystem players (Tarabut, Geidea, Lean Technologies, HyperPay, Tamara, Tabby) are tracked as connector/partner entities under SIG-PATH and SIG-PARTNER.

2. SIGNAL SOURCE SPECIFICATIONS — PER STREAM

SIG-NEWS — NEWS, ANNOUNCEMENTS & PRESS

Source Name	Google News API (Custom Search JSON API)
Endpoint / URL	https://customsearch.googleapis.com/customsearch/v1 — searchType=news
Authentication	API Key + Custom Search Engine ID (cx). Free tier: 100 queries/day. Paid: \$5/1000 queries.
Per-Account Query	Per-account: query="{bank_name}" AND ("digital transformation" OR "lending" OR "technology" OR "fintech" OR "core banking" OR "open banking" OR "SAMA"). SNB example: query="Saudi National Bank" OR "SNB Alahli" AND (digital OR lending OR technology).
Cadence	Hourly for HOT-tier accounts. Every 4 hours for WARM. Daily for COLD.
Signal Emitted	signal_type=NEWS, priority=computed from keyword match depth + source trust.
Per-Account Config	Each account has a query template stored in Rule_Registry with: bank_name variants (English + Arabic), bank domain, subsidiary names. SNB: ["Saudi National Bank", "SNB", "Alahli", "██", "Neo digital bank"]. Al Rajhi: ["Al Rajhi Bank", "██████ ██████████", "Al Rajhi Capital", "EJADA"].
EPIS Profile	E=0.70, P=0.50, I=0.70, S=0.60 — aggregated news; moderate reliability, low proximity (reported not direct).
Notes / Limits	Rate limit: 100 free/day across ALL accounts. At 10 accounts × hourly = 240 queries/day = paid tier required. Budget: ~\$1.20/day = ~\$36/month. Deduplicate against bank newsroom (same event, two sources = one fact with badge rise).

Source Name	Bank Official Newsroom (Web Scrape)
Endpoint / URL	Per-account: SNB: alahli.com/en/about-us/media-center Al Rajhi: alrajhibank.com.sa/en/media Riyadh: riyadbank.com/en/media SAB: sab.com/en/about-us/media Alinma: alinma.com/en/media ANB: anb.com.sa/en/media-center BAJ: baj.com.sa/en/media Albilad: bankalbilad.com/en/media
Authentication	None (public). Scrape via n8n HTTP Request + Cheerio/Puppeteer. Respect robots.txt. Rate limit: 1 request per bank per 30 minutes.
Per-Account Query	No query — full newsroom page scrape. Parse: headline, date, category, body text. Entity-resolve to account. Fingerprint: bank_id + headline_hash + date.
Cadence	Every 30 minutes for HOT accounts. Every 2 hours for WARM. Every 6 hours for COLD.
Signal Emitted	signal_type=NEWS, priority=HIGH (official source). Source trust: 0.85 (bank official, per ICS-SOURCE-001).
Per-Account Config	Each account has: newsroom_url, css_selectors for headline/date/body (varies per bank site), last_scraped_id (dedup). CSS selectors must be maintained — site redesign breaks scraper.
EPIS Profile	E=0.85, P=0.80, I=0.50, S=0.70 — official bank communication; high evidence + proximity, low independence (self-reported).
Notes / Limits	Arabic-language newsrooms exist for most banks (separate URLs). Both EN and AR must be scraped. Arabic NLP parsing needed for entity extraction.

Source Name	Reuters ME / Bloomberg ME (News Wire)
Endpoint / URL	Reuters: reuters.com/news/archive/saudi-arabia — filtered by financial sector. Bloomberg: bloomberg.com/middleeast (paywall). Alternative: Refinitiv Eikon API for Reuters wire.
Authentication	Refinitiv Eikon API: commercial license (\$\$\$). Bloomberg Terminal API: enterprise license. Alternative: Google News API captures Reuters/Bloomberg articles without direct API.
Per-Account Query	Same as Google News API — Reuters/Bloomberg articles surface through Google News. Direct API only if commercial license justifies it.
Cadence	Via Google News API (no separate call needed unless direct feed).
Signal Emitted	signal_type=NEWS, source_trust=0.85 (established news, per ICS-SOURCE-001).
Per-Account Config	Captured via Google News API bank-name queries. No separate per-account config unless direct wire feed.
EPIS Profile	E=0.85, P=0.40, I=0.90, S=0.60 — high independence (journalist-verified), low proximity (external observer).
Notes / Limits	COST DECISION REQUIRED: Direct Reuters/Bloomberg API is expensive (\$500-2000+/month). Google News API captures the same articles with 1-4 hour delay. Recommendation: Google News API sufficient for MVP; direct wire only if sub-hour latency required.

Source Name	Arab News / Saudi Gazette / Argaam (Local Press)
Endpoint / URL	Arab News: arabnews.com (EN). Saudi Gazette: saudigazette.com.sa (EN). Argaam: argaam.com (AR — premier Saudi financial news). Maaal: maaal.com (AR financial).
Authentication	None (public). RSS feeds available: arabnews.com/rss.xml, argaam.com/rss.
Per-Account Query	RSS feed subscription filtered by category (banking, finance, technology). Keyword filter: bank name variants. Argaam is particularly valuable — Saudi financial community reads it before Reuters.
Cadence	RSS poll every 15 minutes for Argaam (most time-sensitive). Every 30 minutes for Arab News. Hourly for others.
Signal Emitted	signal_type=NEWS, priority=MEDIUM-HIGH. Source trust: Argaam 0.80, Arab News 0.75, Saudi Gazette 0.70.
Per-Account Config	RSS feeds are not per-account — they are firehose feeds filtered by bank name keyword matching after ingestion. The per-account config is the keyword filter set (same as Google News query template).
EPIS Profile	E=0.75, P=0.50, I=0.75, S=0.55 — local press; moderate across all dimensions. Argaam higher on proximity (Saudi financial community source).
Notes / Limits	Argaam is Arabic-only. Arabic NLP pipeline required. Argaam often breaks Saudi banking news before English-language outlets — 4-12 hour lead time advantage.

SIG-REG — REGULATORY SIGNALS — SAMA / CMA / TADAWUL

Source Name	SAMA Official Website — Circulars & Directives
Endpoint / URL	https://www.sama.gov.sa/en-US/Laws/Pages/default.aspx (Laws & Regulations) https://www.sama.gov.sa/en-US/News/Pages/default.aspx (News) RSS: sama.gov.sa/en-US/News/RSSfeed.aspx
Authentication	None (public). Web scrape + RSS.
Per-Account Query	No query — full circular/directive listing. Parse: circular number, date, title, category (Banking, Insurance, Finance Companies, Payments). Filter: banking-relevant circulars. Entity-resolve: does this circular affect target accounts specifically (named) or all banks (sector-wide)?
Cadence	Every 2 hours (SAMA publishes irregularly, but compliance-urgency signals are time-critical).
Signal Emitted	signal_type=REGULATORY, priority=CRITICAL (compliance-urgency window opens). Triggers: compliance-urgency modifier in Timing module.
Per-Account Config	SAMA circulars are mostly sector-wide (apply to all banks). Per-account specificity comes from: (a) named bank in circular (rare but high-value), (b) circular topic matching account known strategic initiatives (e.g., open banking circular + account has open banking initiative = HIGH relevance).
EPIS Profile	E=0.95, P=0.90, I=0.95, S=0.80 — highest reliability source in the system (government regulator). Per ICS-SOURCE-001: SAMA Official = 0.95.
Notes / Limits	Source trust: 0.95 (per ICS-SOURCE-001). SAMA circulars create compliance windows with explicit deadlines — these map directly to the Timing module Window Multiplier. Arabic is the authoritative language; English translations may lag.

Source Name	Capital Market Authority (CMA)
Endpoint / URL	https://cma.org.sa/en/RulesRegulations/Pages/default.aspx News: cma.org.sa/en/MediaCenter/Pages/default.aspx
Authentication	None (public). Web scrape.
Per-Account Query	Filter by: banking/financial sector relevance. Most CMA regulations affect capital markets (IPOs, securities) — less directly relevant to Decimal than SAMA, but fintech licensing and sandbox updates are CMA-domain.
Cadence	Daily.
Signal Emitted	signal_type=REGULATORY, priority=MEDIUM (unless fintech licensing — then HIGH).
Per-Account Config	CMA signals are sector-wide. Per-account relevance via topic matching only.
EPIS Profile	E=0.95, P=0.85, I=0.95, S=0.70 — government authority, slightly lower specificity than SAMA for banking.
Notes / Limits	CMA manages the Fintech Experimental Permit (sandbox). New sandbox entrants or graduates are HIGH signal — they indicate fintech ecosystem movement.

Source Name	Tadawul (Saudi Exchange) — Disclosures & Filings
Endpoint / URL	https://www.saudiexchange.sa/wps/portal/saudiexchange/listing/company-disclosures — per ticker. API: iiss.saudiexchange.sa (Tadawul API, commercial).
Authentication	Public web for disclosures page. Tadawul API (iiss.saudiexchange.sa) requires commercial subscription for real-time. Free: disclosure page scrape.
Per-Account Query	Per-account by Tadawul ticker: SNB=1180, Al Rajhi=1120, Riyad=1010, SAB=1060, Alinma=1150, ANB=1080, BAJ=1020, Albilad=1140. Filter: board announcements, financial results, material events.
Cadence	Every 4 hours (disclosures are date-stamped; intraday for material events if API licensed).
Signal Emitted	signal_type=FINANCIAL or REGULATORY depending on disclosure type. Board changes -> SIG-EXEC cross-link. Financial results -> SIG-FIN.
Per-Account Config	Ticker-based. Each account has its Tadawul ticker in the account entity. Disclosure scrape URL: saudiexchange.sa/wps/portal/saudiexchange/listing/company-disclosures?companySymbol={ticker} .
EPIS Profile	E=0.95, P=0.85, I=0.95, S=0.85 — official exchange filings; near-SAMA reliability.
Notes / Limits	Material event disclosures (board changes, M&A, regulatory actions) are the highest-value Tadawul signals. Financial results feed SIG-FIN. Arabic is primary; English translation available for most filings.

SIG-FIN — FINANCIAL POSITION & BUDGET SIGNALS

Source Name	Bank Investor Relations Pages
Endpoint / URL	SNB: alahli.com/en/about-us/investor-relations Al Rajhi: alrajhibank.com.sa/en/investor-relations Riyad: riyadbank.com/en/investor-relations SAB: sab.com/en/investor-relations Alinma: alinma.com/en/investor-relations ANB: anb.com.sa/en/investor-relations
Authentication	None (public). Scrape for new annual reports, quarterly results PDFs, investor presentations.
Per-Account Query	Per-account: scrape IR page for new PDFs (annual report, quarterly results, investor deck). Parse: revenue, net income, IT spend (if disclosed), digital banking KPIs, technology strategy section.
Cadence	Quarterly (aligned with reporting calendar: Q1=Apr, Q2=Jul, Q3=Oct, Annual=Mar). Check weekly for off-cycle publications.
Signal Emitted	signal_type=FINANCIAL, priority=HIGH on publish. Maps to Budget Modifier in Effective Opportunity scoring.
Per-Account Config	Each account has: ir_url, last_annual_report_date, last_quarterly_date, known IT spend (if disclosed). Annual reports contain technology strategy sections that feed EXEC_BELIEF_TIMELINE.
EPIS Profile	E=0.90, P=0.80, I=0.60, S=0.75 — official but self-reported (audited financials are high evidence; strategy commentary is lower independence).
Notes / Limits	Annual reports are the richest single source: they reveal vendor names (SIG-VENDOR), subsidiary activity (SIG-SUBS), strategic initiatives, technology priorities, and sometimes IT spend. Must be parsed in full, not just financial tables.

Source Name	Rating Agencies (Fitch, Moody's, S&P;)
Endpoint / URL	Fitch: fitchratings.com/search?query={bank_name}&type;=issuer Moody's: moody.com/credit-ratings/{entity} S&P: spglobal.com/ratings (paywall)
Authentication	Public for rating summaries. Full reports behind paywall. RSS/alerts available for rating changes.
Per-Account Query	Per-account: monitor for rating changes (upgrade/downgrade/outlook change). Any rating action is a material signal.
Cadence	Daily check for rating actions. Quarterly for outlook reviews.
Signal Emitted	signal_type=FINANCIAL, sub_type=RATING_CHANGE. Upgrade -> budget confidence signal. Downgrade -> risk signal (may slow procurement).
Per-Account Config	Each bank has existing ratings from 1-3 agencies. Track: current_rating, outlook, last_review_date.
EPIS Profile	E=0.90, P=0.30, I=0.95, S=0.70 — highest independence (external expert assessment), lowest proximity (market-level, not account-internal).
Notes / Limits	Rating downgrades can freeze technology procurement for 6-12 months. This is a negative signal that should cap tier or trigger watch mode.

Source Name	MAGNiTT / Crunchbase (Subsidiary Funding)
Endpoint / URL	MAGNiTT: magnitt.com/api (MENA startup DB). Crunchbase: api.crunchbase.com/api/v4 (global).
Authentication	MAGNiTT: API key (startup/enterprise plan). Crunchbase: API key (Basic free = 200 calls/min; Pro = \$29/mo).
Per-Account Query	Per-subsubsidiary: search for funding rounds where org_name matches bank subsidiary (Neo, Emkan, D360, Tiqmo, STC Pay). Crunchbase: GET /organizations/{slug}/funding_rounds. MAGNiTT: search by org name + KSA filter.
Cadence	Weekly scan.
Signal Emitted	signal_type=FUNDING, priority=HIGH if round > \$10M. Maps to budget-availability signal for subsidiary entity.
Per-Account Config	Configured per subsidiary entity, not parent bank. Each subsidiary has: magnitt_slug, crunchbase_slug, last_known_round.
EPIS Profile	Crunchbase: E=0.80 P=0.50 I=0.80 S=0.70. MAGNiTT: E=0.75 P=0.60 I=0.75 S=0.80 (better MENA coverage).
Notes / Limits	OPEN-GAP-SIG-03 gap. Currently not in V2 INT-SIG. Requires governance decision. MAGNiTT has better KSA coverage than Crunchbase for early-stage entities.

SIG-EXEC — EXECUTIVE INTELLIGENCE & DOSSIER

Source Name	LinkedIn Personal Profiles (via Heyreach / Apollo)
Endpoint / URL	Heyreach: app.heyreach.io (LinkedIn automation). Apollo: api.apollo.io/v1/people/search. LinkedIn Sales Navigator: linkedin.com/sales (if licensed).
Authentication	Heyreach: account login + subscription. Apollo: API key (\$49-99/mo). Sales Navigator: enterprise license (\$99+/mo/seat).
Per-Account Query	Per-account: search current_company="{bank_name}" AND seniority=["VP","Director","C-Suite","Partner"]. SNB example: current_company="Saudi National Bank" OR "SNB" AND title contains ["CTO","CDO","CIO","CEO","COO","CXO","Head of Digital","Head of Lending","Head of Technology","VP Technology","SVP"]. Track: name, title, tenure_start, profile_url, recent_posts.
Cadence	Weekly full scan. Event-driven re-check when SIG-NEWS detects executive-related news.
Signal Emitted	signal_type=EXEC_MOVE (new hire/departure), EXEC_PROFILE (dossier update). New C-suite hire -> P1 signal (vendor evaluation window). Departure -> dual signal (vacancy + follow-the-person).
Per-Account Config	Each account has: target_titles (list of C-suite/decision-maker roles to track), known_contacts (from Buying_Committee_Member entities), last_scan_date. The buying committee is per-account — SNB's is different from AI Rajhi's.
EPIS Profile	E=0.60, P=0.90, I=0.30, S=0.60 — self-reported (low independence), but high proximity (the person themselves). Per ICS-SOURCE-001: LinkedIn personal = 0.60.
Notes / Limits	Heyreach rate limits: ~100 profile views/day per LinkedIn account. With 10 target banks × 10-20 execs each = 100-200 profiles. One LinkedIn account insufficient — need 2-3 Heyreach seats or staggered scanning. BAN RISK: excessive scraping triggers LinkedIn restrictions. Circuit breaker per INT-LIN rules.

Source Name	Conference Agendas & Speaker Lists (Named Events)
Endpoint / URL	Money20/20 ME: money2020.com/middle-east (Riyadh, annually in Nov/Dec). Seamless ME: seamless-expo.com (Riyadh, annually). GITEX Global: gitex.com (Dubai, annually Oct). Fintech Saudi Summit: fintechsaudi.com (Riyadh, annually). Sibos: sibos.com (global, annually). Saudi Capital Market Forum: cma.org.sa events. Future Investment Initiative (FII): futureinvestmentinitiative.com (Riyadh, annually Oct). LEAP: oneleap.com (Riyadh, annually Feb).
Authentication	Public (speaker lists and agendas published on event websites). Some require registration for full agenda.
Per-Account Query	Per-event: scrape speaker/panelist list. Match against: (a) target account exec names, (b) target account company names. Capture: speaker name, session title, session abstract, date, time, venue. This is SIG-EXEC grounded material — a captured speech topic IS personalization content.
Cadence	Event-driven: 4-6 weeks before each event (agendas published), day-of (live updates), post-event (recordings/press). Annual calendar: LEAP (Feb), Seamless (Apr/May), FII (Oct), GITEX (Oct), Money20/20 (Nov/Dec), Fintech Saudi (varies), Sibos (varies).
Signal Emitted	signal_type=EVENT, cross-link to SIG-EXEC (speech material) + Timing module (event proximity = engagement window). If target exec is speaking -> HIGH priority timing signal.
Per-Account Config	Match speaker names against known buying committee for each account. Unknown speakers at target banks are also captured — potential new contacts.
EPIS Profile	E=0.80, P=0.70, I=0.70, S=0.75 — event agendas are official; speech content is grounded, sourced, and dated.
Notes / Limits	CRITICAL EVENTS for KSA banking: (1) LEAP — Saudi tech mega-event, CxOs attend and announce. (2) Money20/20 ME — fintech-specific, Dr. Al-Furaih (SNB NEO CEO) presented in 2024. (3) FII — "Davos in the Desert," CEOs attend. (4) Seamless — payments/commerce, operational leaders attend. (5) Fintech Saudi — regulator-adjacent, compliance and innovation leaders. Build annual event calendar with scrape schedule per event.

Source Name	Bank Annual Reports — Executive Section (PDF Parse)
Endpoint / URL	Same as SIG-FIN IR pages. Specific pages: Board of Directors listing, Executive Management listing, Corporate Governance section.
Authentication	None (public PDFs).
Per-Account Query	Parse annual report PDF for: Board of Directors names + committee assignments, Executive Management names + titles, organizational chart (if included), executive bios. Compare against prior year for: new appointments, departures, title changes.
Cadence	Annually (on annual report publication, typically Q1 for prior-year report).
Signal Emitted	signal_type=EXEC_PROFILE (bulk dossier refresh), EXEC_MOVE (if changes detected vs. prior year).
Per-Account Config	Annual report PDF URL stored per account. Parser extracts structured data from known sections. Each bank's annual report has different formatting — parser templates per bank.
EPIS Profile	E=0.90, P=0.80, I=0.60, S=0.85 — audited document; names/titles are verified. Highest-confidence source for executive roster.
Notes / Limits	Annual reports are the AUTHORITATIVE source for executive rosters (per SIG-SUBS rules). LinkedIn may show stale titles; the annual report is the verification benchmark.

SIG-HIRE [NEW] — HIRING & JOB POSTING INTENT SIGNALS (OPEN-GAP-SIG-01)

Source Name	LinkedIn Jobs (via LinkedIn API / Heyreach)
Endpoint / URL	linkedin.com/jobs/search/?keywords={bank_name}&location;=Saudi+Arabia API: api.linkedin.com/v2/jobSearch (requires LinkedIn Recruiter or Marketing API approval). Alternative: scrape via Heyreach or Phantombuster.
Authentication	LinkedIn Jobs API: OAuth 2.0 + approved app (restrictive — LinkedIn limits job scraping). Heyreach: LinkedIn account-based scraping. Phantombuster: headless browser automation.
Per-Account Query	Per-account: company_id={linkedin_company_id} AND job function IN ["Information Technology","Engineering","Product Management","Digital"]. SNB: company=snbalahli, keywords=["lending","digital","core banking","open banking","onboarding","origination","API","microservices","cloud"]. Parse: job title, department, requirements (tech stack!), seniority, posting date.
Cadence	Daily for all target accounts.
Signal Emitted	signal_type=HIRING_INTENT, priority computed from: (a) title relevance to Decimal products (Head of Digital Lending = P1), (b) seniority (C-suite hire = SIG-EXEC cross-link), (c) tech stack in requirements (Temenos/Backbase/FIS mentioned = SIG-VENDOR cross-link).
Per-Account Config	Each account has: linkedin_company_id, relevant_keyword_set (tuned to Decimal's product portfolio), last_job_scan_date, active_postings_count. The keyword set is the critical config — it determines what counts as a buying-intent signal vs. generic hiring.
EPIS Profile	E=0.80, P=0.70, I=0.80, S=0.90 — verified posting (someone authorized it), high specificity (role + requirements are explicit intent).
Notes / Limits	THIS IS THE HIGHEST-VALUE MISSING SOURCE. A bank posting "Head of Digital Lending Platform" with "experience with Temenos/Backbase/Finacle" in requirements is: (a) confirmed budget (role approved), (b) confirmed timeline (hiring = 3-6 month procurement horizon), (c) tech stack revelation (SIG-VENDOR), (d) contact identification (the hire becomes a buying committee member). LinkedIn rate limits apply — circuit breaker per INT-LIN.

Source Name	Bayt.com (MENA Job Board)
Endpoint / URL	bayt.com/en/saudi-arabia/jobs/?keyword={bank_name} No public API — web scrape required.
Authentication	None for public listings. Scrape via n8n HTTP Request + HTML parser.
Per-Account Query	Per-account: search company="{bank_name}" AND country="Saudi Arabia" AND sector="Banking/Finance". Filter by: IT, Technology, Digital, Engineering departments.
Cadence	Daily.
Signal Emitted	signal_type=HIRING_INTENT. Lower priority than LinkedIn (less tech-specific postings). Cross-reference with LinkedIn postings for dedup.
Per-Account Config	Company name on Bayt may differ from LinkedIn (e.g., "The Saudi National Bank" vs "SNB"). Mapping table required.
EPIS Profile	E=0.70, P=0.50, I=0.70, S=0.80 — aggregated job board; postings are real but may lag LinkedIn.
Notes / Limits	Bayt is the dominant job board in GCC. Some roles appear on Bayt but NOT LinkedIn (especially mid-level operational roles). Dual-source coverage prevents blind spots.

Source Name	Bank Career Pages (Direct Scrape)
Endpoint / URL	Per-account: SNB: alahli.com/careers Al Rajhi: alrajhibank.com.sa/en/careers Riyadh: riyadbank.com/en/careers SAB: sab.com/en/careers Alinma: alinma.com/en/careers ANB: anb.com.sa/en/careers. Some banks use Workday/Taleo portals: workday.com/{company}/jobs, oracle.com/taleo/{company}.
Authentication	None (public). Some career portals are JavaScript-rendered (require headless browser).
Per-Account Query	Per-account: scrape all open positions. Filter by department/category: Technology, Digital, IT, Engineering, Product. Parse: title, department, location, posting date, description (for tech stack keywords).
Cadence	Daily for HOT accounts. Every 3 days for WARM/COLD.
Signal Emitted	signal_type=HIRING_INTENT. Bank career pages are the AUTHORITATIVE source (postings may appear here before LinkedIn). Cross-reference for dedup.
Per-Account Config	Each account has: careers_url, portal_type (static HTML / Workday / Taleo / iCIMS), css_selectors or API endpoint, last_scraped_date.
EPIS Profile	E=0.85, P=0.80, I=0.60, S=0.85 — direct from the bank (high proximity, lower independence). Highest specificity — full job descriptions available.
Notes / Limits	Career page scraping is fragile — portal redesigns break scrapers. Headless browser (Puppeteer/Playwright) needed for JavaScript-rendered portals (Workday, Taleo). Maintenance cost is real but signal value justifies it.

Source Name	GulfTalent (Premium MENA Executive Recruitment)
Endpoint / URL	gulftalent.com/jobs/banking-saudi-arabia No public API.
Authentication	Public listing view. Premium access for full descriptions (subscription required).
Per-Account Query	Per-sector: search industry="Banking" AND country="Saudi Arabia" AND function IN ["IT","Technology","Digital"]. Match company names to target accounts.
Cadence	Weekly (GulfTalent is a secondary source; LinkedIn + Bayt + career pages cover primary).
Signal Emitted	signal_type=HIRING_INTENT, priority=MEDIUM. GulfTalent skews toward senior/expat roles — useful for detecting international hires (which correlate with transformation initiatives).
Per-Account Config	Company name matching against target account list.
EPIS Profile	E=0.70, P=0.40, I=0.70, S=0.75 — aggregator; less timely than direct sources.
Notes / Limits	GulfTalent's niche is expat executive recruitment. A KSA bank hiring a "Head of Digital Banking" from London/Singapore via GulfTalent is a stronger transformation signal than a local hire — it suggests international best-practice mandate.

SIG-VENDOR — VENDOR RELATIONSHIPS & COMPETITIVE INTELLIGENCE

Source Name	Vendor Company Websites — Case Studies & Press Releases
Endpoint / URL	Backbase: backbase.com/customers + backbase.com/news Temenos: temenos.com/news + temenos.com/customer-stories FIS: fisglobal.com/en/about-us/news Finastra: finastra.com/news-events Thought Machine: thoughtmachine.net/customers Mambu: mambu.com/customers Efigence: efigence.com/case-studies.
Authentication	None (public). Web scrape.
Per-Account Query	Per-vendor: scrape case studies and press releases. Filter by: (a) Saudi Arabia / KSA / MENA geography, (b) target bank names. A case study naming a target bank = CONFIRMED vendor relationship.
Cadence	Monthly scan of all monitored vendor sites. Event-driven: if SIG-NEWS detects vendor-bank partnership announcement.
Signal Emitted	signal_type=VENDOR_RELATIONSHIP, sub_type=CONFIRMED or INFERRED. Confirmed: bank named in case study. Inferred: job posting mentions vendor tech stack.
Per-Account Config	Per-account vendor registry: known vendors, relationship status (incumbent/partner/evaluating/rumored), source of knowledge, last_verified_date. Exclusivity discipline: Backbase=SNB only, Efigence=Riyad only, per spec.
EPIS Profile	E=0.85, P=0.50, I=0.40, S=0.90 — vendor-sourced (low independence — vendor has incentive to publicize), high specificity (named client, named product).
Notes / Limits	COMPETITIVE INTELLIGENCE is the primary purpose. If Backbase publishes an Al Rajhi case study, that is a HIGH-urgency signal for Decimal — Al Rajhi's vendor evaluation for digital banking may be closing. Per SIG-VENDOR-01 rules, each bank's vendor registry is per-account and exclusivity-bound.

Source Name	Job Postings — Tech Stack Inference (Cross-link from SIG-HIRE)
Endpoint / URL	Same as SIG-HIRE sources (LinkedIn Jobs, Bayt, bank career pages).
Authentication	Same as SIG-HIRE.
Per-Account Query	Parse job descriptions for technology keywords: "Temenos T24", "Backbase", "FIS Modern Banking Platform", "Finastra Fusion", "Thought Machine Vault", "Mambu", "Oracle FLEXCUBE", "SAP Banking", "Infosys Finacle". A bank hiring for "Temenos T24 Developer" confirms Temenos as incumbent.
Cadence	Piggybacks on SIG-HIRE daily scan — no additional source calls. Tech stack extraction is a parsing step on SIG-HIRE output.
Signal Emitted	signal_type=VENDOR_INFERRED, confidence=MEDIUM (job posting is not a vendor confirmation — it could be legacy maintenance or new evaluation).
Per-Account Config	Tech stack keywords stored per account. When new vendor keyword detected in job posting, flag for human verification before updating vendor registry.
EPIS Profile	E=0.70, P=0.60, I=0.70, S=0.85 — inferred (not vendor-confirmed), but highly specific (named technology in job requirements).
Notes / Limits	This is where SIG-HIRE and SIG-VENDOR intersect. A single job posting can emit both HIRING_INTENT (they're investing in digital lending) and VENDOR_INFERRED (they use Temenos T24). The parsing pipeline must emit both signal types from one source event.

SIG-TENDER [NEW] — PROCUREMENT & RFP SIGNALS (OPEN-GAP-SIG-02)

Source Name	Etimad (Saudi Government Procurement Platform)
Endpoint / URL	https://etimad.sa — Government procurement portal. Banks with government ownership (SNB, Al Rajhi Capital for government mandates) may publish tenders here.
Authentication	Public tender listing view (no auth for browsing). Bid submission requires registered vendor account.
Per-Account Query	Search: category="Information Technology" OR "Banking Technology" OR "Digital Transformation" AND entity matches target accounts or government financial entities. Monitor new tender postings.
Cadence	Daily scan.
Signal Emitted	signal_type=TENDER, priority=CRITICAL (active procurement). Triggers opportunity state machine transition to ACTIVE PROCUREMENT.
Per-Account Config	Government-owned banks (SNB has PIF ownership) may route through Etimad. Private banks use their own procurement — Etimad coverage is partial.
EPIS Profile	E=0.95, P=0.90, I=0.95, S=0.95 — government procurement platform; near-SAMA reliability. Highest possible signal confidence.
Notes / Limits	GOVERNANCE DECISION REQUIRED: Etimad coverage of private bank tenders is limited. Most Saudi banks run procurement through internal portals or consulting intermediaries (Deloitte, EY, McKinsey). The manual-input path (BD team enters known RFPs as Signal entities) may be more reliable than automated scraping for non-government entities.

Source Name	Manual RFP Input (BD Team Feed)
Endpoint / URL	Internal: FastAPI endpoint POST /api/v1/signals with signal_type=TENDER. BD team enters RFP details when received via network intelligence.
Authentication	Internal API auth (JWT token, Decimal employee only).
Per-Account Query	Structured input: bank_name, rfp_title, scope_description, deadline, contact_person, source_of_knowledge (who told us), estimated_value.
Cadence	As received (event-driven).
Signal Emitted	signal_type=TENDER, priority=CRITICAL, human_sourced=true.
Per-Account Config	Per the account the RFP is from. Links to existing account entity.
EPIS Profile	E=0.85, P=0.90, I=0.30, S=0.90 — human-sourced (high proximity, low independence — Decimal employee is the source). Per ICS-SOURCE-001: verified manual BD = 0.85.

Notes / Limits	P1 implementation path. This is trivially buildable — a form/API endpoint for the BD team. Automation (Etimad scraping, consulting-firm feed) is P2. The most common path for RFP awareness in KSA banking is network intelligence (someone tells you), not published tenders.
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SIG-EVENT — EVENTS, CONFERENCES & SPEAKING SIGNALS

Source Name	Named Conference Calendar — KSA Banking & Fintech
Endpoint / URL	LEAP: oneleap.com (Riyadh, Feb) Seamless ME: seamless-expo.com (Riyadh, Apr/May) Future Investment Initiative (FII): futureinvestmentinitiative.com (Riyadh, Oct) GITEX Global: gitex.com (Dubai, Oct) Money20/20 ME: money2020.com/middle-east (Riyadh, Nov/Dec) Fintech Saudi Summit: fintechsaudi.com (Riyadh, varies) Saudi Capital Market Forum: cma.org.sa (Riyadh, annually) Sibos: sibos.com (global, annually) World Economic Forum Davos: weforum.org (Jan) MENA Fintech Association Summit: menafintech.org (varies) Arab Banking Conference: abconference.com (varies).
Authentication	Public (speaker lists published on event websites). Some require free registration for full agenda.
Per-Account Query	Per-event: (a) scrape published agenda / speaker list, (b) match speaker names against target account buying committees + C-suite, (c) capture session titles and abstracts as grounded personalization material (SIG-EXEC-02), (d) capture sponsorship/exhibitor lists (SIG-VENDOR signal if competitor is sponsoring).
Cadence	Annual calendar with per-event scrape schedule: 6 weeks before (agenda published), 2 weeks before (final speakers confirmed), day-of (live social monitoring), 1 week after (press coverage of presentations).
Signal Emitted	signal_type=EVENT, priority=HIGH if target exec is speaking. Cross-links: SIG-EXEC (speech capture), Timing module (event proximity = engagement window — 2 weeks before to 2 weeks after).
Per-Account Config	Match against known buying committee + C-suite names. Unknown names at target banks are flagged as potential new contacts for SIG-EXEC update.
EPIS Profile	E=0.80, P=0.70, I=0.70, S=0.75 — event agendas are official; content is grounded and dated.
Notes / Limits	CRITICAL TIMING SIGNALS: (1) Dr. Al-Furaih (SNB NEO CEO) presented at Money20/20 ME 2024 — engagement window around this event was the optimal approach moment. (2) FII is where Saudi banking CEOs make strategic announcements. (3) LEAP is where technology partnerships are announced. The event calendar is a first-class timing instrument, not just a data source.

Source Name	LinkedIn Company Page — Events Tab
Endpoint / URL	Per-account: linkedin.com/company/{company_slug}/events. Per-exec: linkedin.com/in/{profile}/recent-activity (posts about attending events).
Authentication	Heyreach / LinkedIn account.
Per-Account Query	Per-account: check company events tab for hosted/sponsored events. Per-exec: monitor posts for "excited to speak at" / "attending" / "just presented at" signals.
Cadence	Weekly scan of company event tabs. Daily monitoring of exec posts for event mentions.
Signal Emitted	signal_type=EVENT, sub_type=ATTENDANCE or SPEAKING or HOSTING.
Per-Account Config	Linked to known exec profiles per account.
EPIS Profile	E=0.60, P=0.90, I=0.30, S=0.60 — self-reported (low independence), but high proximity.
Notes / Limits	Exec LinkedIn posts about attending events are the EARLIEST signal that someone will be at a specific event — often before the official agenda is published. This enables pre-event outreach scheduling.

SIG-SUBS — SUBSIDIARY & VENTURE ACTIVITY

Source Name	Bank Annual Report — Subsidiary Listings (Authoritative)
Endpoint / URL	Same as SIG-FIN IR pages. Annual report corporate governance section lists subsidiaries with ownership % and key officers.
Authentication	None (public PDF).
Per-Account Query	Parse annual report for: subsidiary names, ownership percentages, subsidiary board members, subsidiary financial highlights. Compare vs. prior year for: new subsidiaries, dissolved subsidiaries, leadership changes.
Cadence	Annually (on annual report publication).
Signal Emitted	signal_type=SUBSIDIARY_UPDATE. New subsidiary = expansion signal. Leadership change = SIG-EXEC cross-link.
Per-Account Config	Known subsidiaries per bank: SNB -> [ITQAN, Neo, SNB Capital, Alahli Takaful, Quick Pay]. Al Rajhi -> [EJADA, Al Rajhi Capital, Al Rajhi Takaful, Emkan Finance]. Riyad -> [Riyad Capital]. Each subsidiary has its own entity with: name, website, linkedin_page, known_officers, activity_signals.
EPIS Profile	E=0.90, P=0.80, I=0.60, S=0.85 — audited document. Authoritative for subsidiary roster (per SIG-SUBS-01).
Notes / Limits	Annual reports are the AUTHORITATIVE source for subsidiaries per spec. IT subsidiaries are critical gatekeepers — ITQAN (SNB), EJADA (Al Rajhi) must be routed through before C-suite per the gatekeeper-sequencing tradecraft rule.

Source Name	Subsidiary Websites & LinkedIn Pages (Ongoing Activity)
Endpoint / URL	Per-subsidiary: ITQAN: itqan.com.sa + linkedin.com/company/itqan. EJADA: ejada.com + linkedin.com/company/ejada-systems. Neo: neo.sa (SNB digital bank). Emkan: emkan.com.sa. D360: d360bank.com.
Authentication	None (public).
Per-Account Query	Monitor subsidiary websites for: news, new products, partnerships, job postings. Monitor LinkedIn for: company updates, employee count changes, new hires.
Cadence	Weekly.
Signal Emitted	signal_type=SUBSIDIARY_ACTIVITY. Job postings -> cross-link to SIG-HIRE. News -> cross-link to SIG-NEWS.
Per-Account Config	Each subsidiary has its own monitoring config: website_url, linkedin_url, last_scraped_date.
EPIS Profile	E=0.75, P=0.70, I=0.50, S=0.70 — subsidiary self-reporting; moderate across all dimensions.

Notes / Limits	Subsidiary employee count changes on LinkedIn are a leading indicator of investment/divestment. A 20%+ headcount jump at ITQAN signals SNB is ramping internal technology capacity — potential negative signal (build vs. buy) or positive (they're investing and need partners).
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SIG-PATH — CONNECTION-TO-DECIMAL RESOLUTION

Source Name	LinkedIn Graph (Decimal Employee Network Ingestion)
Endpoint / URL	linkedin.com/in/{decimal_employee} — connections list. Heyreach: automated connection-degree checking. Apollo: api.apollo.io/v1/people/match.
Authentication	Heyreach / LinkedIn accounts for Decimal employees. Apollo API.
Per-Account Query	For each (Decimal employee, target account contact) pair: determine LinkedIn connection degree (1st, 2nd, 3rd+). For 2nd-degree: identify mutual connections. For 3rd-degree+: identify bridge nodes. This produces the P1-P5 priority ladder per SIG-PATH rules.
Cadence	Continuous (re-check on new contact discovery or Decimal employee connection changes). Full re-resolution weekly.
Signal Emitted	signal_type=CONNECTION_PATH, priority=P1 (1st degree) through P5 (cold). ACCESS_STRENGTH computed per path.
Per-Account Config	Fully per-account and per-contact. The connection matrix is: every Decimal employee × every target account contact. SNB example: Decimal team (N employees) × SNB buying committee (20+ contacts) = N×20 connection checks.
EPIS Profile	E=0.70, P=0.80, I=0.60, S=0.80 — LinkedIn connections are real but connection ≠ relationship strength.
Notes / Limits	This is the headline stream. Per spec: "a signal with no warm path is unactionable." The connection resolution must run against the FULL Decimal employee network, not just the BD team. Engineering, product, and leadership connections matter — the CTO may know someone the BD team doesn't.

Source Name	Generalized Connector Registry (Ecosystem Partners)
Endpoint / URL	Internal database: connectors table in PostgreSQL. Populated from BD intelligence research (SNB/Riyad/AI Rajhi dossiers already completed).
Authentication	Internal (PostgreSQL direct).
Per-Account Query	Per-account: which connector organizations (KPMG, Mastercard, Backbase, Tarabut, Geidea, ITC Infotech, LinkedIn Direct) have mapped paths to target account decision-makers? Connector registry stores: org_name, contact_person, contact_details, path_to_target (which target exec they reach), relationship_strength, last_validated_date.
Cadence	Connector paths updated on: (a) new BD research completion, (b) executive move detected (SIG-EXEC), (c) quarterly re-validation.
Signal Emitted	signal_type=CONNECTION_PATH, sub_type=CONNECTOR_BRIDGE. Priority P3-P4 per SIG-PATH ladder.
Per-Account Config	Fully per-account. SNB has 8 connector paths mapped (ITC Infotech, Backbase, Tarabut, Geidea, KPMG, Mastercard, LinkedIn Direct, ITQAN). Each other bank needs equivalent mapping.
EPIS Profile	E=0.85, P=0.70, I=0.50, S=0.80 — human-researched (high evidence), relationship strength may be stale.
Notes / Limits	The SNB connector map is the reference model — 8 connector paths, 40+ named contacts, all pathways to C-suite mapped. This level of depth must be replicated per target account. BD intelligence research (the skill that produced the SNB/Riyad dossiers) feeds this registry.

SIG-PARTNER [NEW] — MOU & PARTNERSHIP CLASSIFICATION (OPEN-GAP-SIG-06)

Source Name	SIG-NEWS Output — Reclassification Layer
Endpoint / URL	Internal: classification applied to SIG-NEWS signal output. No new external source — this is a downstream processing step.
Authentication	Internal (pipeline processing).
Per-Account Query	For each SIG-NEWS signal, run partnership classifier: (a) Does the news mention a vendor/partner signing? (b) Is the partner a Decimal competitor (Backbase, Temenos, FIS, Finastra, Mambu, Thought Machine)? -> COMPETITIVE_CLOSURE. (c) Is the partner complementary to Decimal (open banking middleware, payment processors, cloud providers)? -> INTEGRATION_OPPORTUNITY. (d) Is it a regulatory/government partnership? -> COMPLIANCE_ALIGNMENT.
Cadence	Real-time (applied to every SIG-NEWS signal as it enters the pipeline).
Signal Emitted	signal_type=PARTNERSHIP, sub_type=COMPETITIVE_CLOSURE INTEGRATION_OPPORTUNITY COMPLIANCE_ALIGNMENT. COMPETITIVE_CLOSURE triggers urgency escalation in scoring.
Per-Account Config	Per-account: which partnerships are detected for which bank. Competitive vendor registry (Backbase=competitor, Tarabut=complementary, KPMG=neutral/advisor) is a system-wide config, but detected partnerships are per-account facts.
EPIS Profile	Inherits EPIS from underlying SIG-NEWS source. Classification adds interpretation layer — classification confidence is separate from source confidence.
Notes / Limits	Requires: (a) competitive vendor list (who competes with Decimal), (b) complementary partner list (who complements Decimal), (c) classification rules for ambiguous cases (a bank partnering with AWS — is that competitive or complementary?). Classification logic lives in the spec, not in n8n (per ORCH-NOLOGIC-001).

Source Name	Fintech Saudi Ecosystem Announcements
Endpoint / URL	fintechsaudi.com/news + fintechsaudi.com/ecosystem (lists all licensed/sandbox fintechs and their bank partnerships).
Authentication	None (public).
Per-Account Query	Monitor for: new fintech-bank partnerships announced, sandbox graduates partnering with banks, ecosystem map updates.
Cadence	Weekly.
Signal Emitted	signal_type=PARTNERSHIP, priority depends on partner identity.
Per-Account Config	Match bank names against target accounts.
EPIS Profile	E=0.80, P=0.60, I=0.80, S=0.70 — semi-official (Fintech Saudi is SAMA-adjacent but not SAMA itself).
Notes / Limits	Fintech Saudi is the ecosystem orchestrator. Their partnership announcements often precede official press releases by the banks themselves.

INT-ENR — ENRICHMENT — CONTACT RESOLUTION & VERIFICATION

Source Name	Apollo.io (Primary Enrichment — First in Chain)
Endpoint / URL	api.apollo.io/v1/people/search api.apollo.io/v1/people/match api.apollo.io/v1/organizations/enrich
Authentication	API Key. Plans: Free (50 credits/mo), Basic (\$49/mo, 5000 credits), Professional (\$99/mo, unlimited). Each person lookup = 1 credit.
Per-Account Query	Per-contact: POST /v1/people/match with: first_name, last_name, organization_name, domain. Returns: verified_email, phone, title, linkedin_url, seniority. Per-org: POST /v1/organizations/enrich with domain. Returns: employee_count, industry, technologies_used, revenue_range.
Cadence	On-demand: triggered by M3 enrichment workflow when account qualifies (HOT/WARM + path-reachable). Re-enrichment on freshness TTL expiry (per INT-ENR-05).
Signal Emitted	Produces Enrichment entity, not Signal. Feeds contact verification, buying committee construction, and reachability determination.
Per-Account Config	Per-contact enrichment. Apollo is the FIRST in the fallback chain (INT-ENR-01: Apollo -> Clearbit -> Clay). Stops at first verified result.
EPIS Profile	E=0.70, P=0.60, I=0.70, S=0.80 (per coherence audit EPIS-ENRICH-001 recommendation).
Notes / Limits	Verified-email-only is NON-NEGOTIABLE (INT-ENR-02). Unverified emails are rejected — they damage sender reputation (INT-EML dependency). Budget: at Professional tier (\$99/mo), unlimited enrichment covers the full target universe.

Source Name	Clearbit (Secondary Enrichment — Firmographic)
Endpoint / URL	api.clearbit.com/v2/people/find api.clearbit.com/v2/companies/find
Authentication	API Key. Pricing: usage-based (\$99-999+/mo depending on volume).
Per-Account Query	Per-contact: GET /v2/people/find?email={email} OR /v2/people/find?domain={domain}&name;={name}. Returns: verified_email, title, seniority, company details. Per-org: GET /v2/companies/find?domain={domain}. Returns: employee_count, tech_stack, revenue, industry_tags.
Cadence	On-demand: triggered when Apollo fails to return verified result (fallback chain position 2).
Signal Emitted	Produces Enrichment entity. Clearbit's tech_stack field feeds SIG-VENDOR inference.
Per-Account Config	Per-contact/per-org. Clearbit is the SECOND in the fallback chain. Called only when Apollo returns no verified email.
EPIS Profile	E=0.80, P=0.50, I=0.90, S=0.70 (per coherence audit EPIS-ENRICH-001 recommendation). Highest independence — third-party compiled data.
Notes / Limits	Clearbit's technology stack detection (tech_stack field) is a valuable SIG-VENDOR input — it identifies what tools a company uses based on web footprint analysis. This can reveal vendor relationships not visible through other sources.

Source Name	Clay (Tertiary Enrichment — Workflow Orchestration)
Endpoint / URL	app.clay.com / Clay API (waterfall enrichment across 75+ providers).
Authentication	Clay account + API key. Pricing: \$149-349/mo depending on credits.
Per-Account Query	Clay acts as a meta-enrichment layer — it waterfalls across multiple providers (Hunter.io, Snov.io, RocketReach, etc.) to find verified emails when Apollo and Clearbit fail. Per-contact: input name + company, Clay tries multiple providers sequentially.
Cadence	On-demand: triggered when both Apollo AND Clearbit fail (fallback chain position 3).
Signal Emitted	Produces Enrichment entity. Clay inherits EPIS from whichever upstream source produced the result.
Per-Account Config	Per-contact. Third in chain. Clay also offers LinkedIn profile enrichment and company research.
EPIS Profile	Inherits from upstream source. Clay itself adds no independent evidence — it's a routing layer.
Notes / Limits	Clay's value is coverage breadth, not depth. For KSA banking contacts, Apollo typically has the best coverage. Clay is the safety net for contacts that fall through both Apollo and Clearbit — particularly useful for subsidiary contacts (ITQAN, EJADA) that may not be well-covered by Apollo/Clearbit.

Source Name	LinkedIn (Parallel Enrichment — Activity & Relationship)
Endpoint / URL	Via Heyreach (linkedin.com automation) or Sales Navigator API.
Authentication	Heyreach subscription + LinkedIn account(s). Sales Navigator: enterprise license.
Per-Account Query	Per-contact: profile view for title verification, recent activity (posts, comments, shares), connection count, mutual connections. LinkedIn is NOT in the email fallback chain — it provides role verification and activity intelligence.
Cadence	Weekly for active buying committee members. Monthly for broader exec roster.
Signal Emitted	Produces: role verification (cross-reference with Apollo/Clearbit title), activity signals (recent posts about digital transformation = engagement timing signal), connection data (feeds SIG-PATH).
Per-Account Config	Per-contact linked to per-account buying committee.
EPIS Profile	E=0.60, P=0.90, I=0.30, S=0.60 (per ICS-SOURCE-001: LinkedIn personal = 0.60). Self-reported — use for proximity, not independence.
Notes / Limits	LinkedIn is the ONLY source for some KSA banking contacts — particularly for verifying Arabic name variants and current titles. INT-LIN rules (Heyreach platform-policy limits, ban-risk circuit breaker) apply strictly. Daily profile view limits: ~100/day per LinkedIn account.

SIG-SOCIAL [NEW / P2] — TWITTER/X & SOCIAL MEDIA SIGNALS (OPEN-GAP-SIG-04)

Source Name	Twitter/X — Executive Account Monitoring
Endpoint / URL	x.com (formerly twitter.com). API: api.x.com/2/tweets/search/recent (X API v2).
Authentication	X API: Basic (\$100/mo, 10k tweets/mo read), Pro (\$5000/mo, 1M tweets/mo). Free tier: 1500 tweets/mo read (insufficient). Alternative: social listening tools (Brandwatch, Sprout Social, Meltwater) that include X data.
Per-Account Query	Per-exec: monitor named exec Twitter accounts for tweets about: digital transformation, fintech, open banking, technology, vendor mentions, event attendance. Per-bank: monitor bank official accounts + hashtags (#SNBAIahli, #AIRajhiBank). Saudi banking hashtags: #SaudiFintech, #OpenBanking, #Vision2030, #SAMA.
Cadence	Daily (API) or real-time (streaming API, Pro tier only).
Signal Emitted	signal_type=SOCIAL, feeds: EXEC_BELIEF_TIMELINE (reasoning stream), SIG-EXEC (grounded material if exec tweets about priorities), SIG-EVENT (event attendance detection).
Per-Account Config	Per-exec Twitter handles mapped to account buying committees. Many KSA banking execs have active Twitter presence — especially younger CxOs. Mapping required: exec_name -> twitter_handle.
EPIS Profile	E=0.50, P=0.90, I=0.30, S=0.40 — lowest independence in the system (self-reported, unverified, informal), but highest proximity (direct from the person, unfiltered).
Notes / Limits	P2 DEFERRAL RECOMMENDED. X API cost (\$100-5000/mo) + low EPIS profile + API instability post-2023 = marginal value for MVP. Saudi Twitter IS culturally significant (Saudi Arabia has one of the highest Twitter penetration rates globally), but the signal-to-noise ratio is low and the cost is high. Revisit when the engine is in production and exec Twitter handles are mapped.

INT-EML + INT-LIN + INT-CRM — OUTREACH & ENGAGEMENT FEEDBACK SOURCES

Source Name	Email Infrastructure (Instantly / Custom SMTP)
Endpoint / URL	Instantly: instantly.ai API. Custom: SMTP server with SPF/DKIM/DMARC configured.
Authentication	Instantly API key. SMTP credentials (Decimal domain).
Per-Account Query	Track per-email: delivered, opened, clicked, replied, bounced, unsubscribed, spam-reported. Per INT-EML rules: SPF/DKIM/DMARC must be configured. Warmup gating applies (new domains cannot send at volume).
Cadence	Real-time (webhook-based). Engagement events flow back immediately.
Signal Emitted	signal_type=ENGAGEMENT, sub_type=EMAIL_OPEN EMAIL_REPLY EMAIL_BOUNCE. Reply = highest-value engagement signal (validates understanding, per ICS-DECAY-002).
Per-Account Config	Per-contact engagement history. Feeds fatigue detection, sentiment classification, and ICS recalculation.
EPIS Profile	E=0.90, P=0.95, I=0.80, S=0.90 — direct behavioral evidence (they opened/replied). Highest proximity possible.
Notes / Limits	Email deliverability is a PREREQUISITE, not just a signal source. Bounced emails degrade sender reputation (domain-level damage affecting ALL outreach). INT-EML warmup gating rules are non-negotiable.

Source Name	Heyreach (LinkedIn Automation)
Endpoint / URL	app.heyreach.io — connection requests, messages, InMails.
Authentication	Heyreach account + LinkedIn account(s).
Per-Account Query	Track per-action: connection_accepted, message_read, message_replied, profile_viewed, post_liked. Heyreach provides event webhooks for these.
Cadence	Real-time (webhook-based).
Signal Emitted	signal_type=ENGAGEMENT, sub_type=LINKEDIN_ACCEPT LINKEDIN_REPLY LINKEDIN_VIEW.
Per-Account Config	Per-contact. LinkedIn engagement is often the FIRST touchpoint (before email). Connection acceptance is a gating event for follow-up messaging.
EPIS Profile	E=0.85, P=0.90, I=0.70, S=0.85 — direct behavioral evidence on LinkedIn platform.
Notes / Limits	INT-LIN circuit breaker rules apply: if Heyreach triggers LinkedIn restrictions, ALL LinkedIn automation must pause across all accounts until the restriction lifts. Ban risk management is per INT-LIN-01 through INT-LIN-07.

Source Name	HubSpot CRM (Bidirectional Sync)
Endpoint / URL	api.hubapi.com/crm/v3/ — contacts, companies, deals, engagements.
Authentication	HubSpot API key or OAuth2 (private app).
Per-Account Query	Bidirectional per INT-CRM rules (COH-F01 resolution): PostgreSQL is SOT for scores/tiers/signals; HubSpot is SOT for deal stage/amount/owner. Sync: push scores/tiers TO HubSpot, pull deal stage/owner FROM HubSpot.
Cadence	Per INT-CRM: idempotent sync every 15 minutes. Event-driven on deal stage changes (webhook).
Signal Emitted	HubSpot is a CONSUMER of engine intelligence, not primarily a signal source. Exception: manual sales rep notes and deal-stage updates feed back as human-sourced signals.
Per-Account Config	Per-account and per-contact CRM records. Account entity maps to HubSpot company. Contact maps to HubSpot contact. Opportunity maps to HubSpot deal.
EPIS Profile	Human-entered data: E=0.80 P=0.85 I=0.30 S=0.70 (sales rep is biased but proximate). Deal stage: SOT by definition.
Notes / Limits	Per COH-F01 resolution: if a sales rep manually changes a field in HubSpot, that field is locked from engine overwrite for 72 hours. Human override flag prevents data oscillation.

Source Name	Metabase (Analytics — Read Only)
Endpoint / URL	Internal Metabase instance. Read-replica of PostgreSQL.

Authentication	Internal (Metabase admin credentials).
Per-Account Query	Metabase is a CONSUMER, not a source. It reads from the PostgreSQL read-replica for dashboards and reporting. Per INT-ANL rules: NO write-back from Metabase to the engine. Analytics cannot create feedback loops.
Cadence	N/A (read-only).
Signal Emitted	None (observation layer only).
Per-Account Config	N/A.
EPIS Profile	N/A — not a signal source.
Notes / Limits	Metabase is the observability layer. It does not produce signals. Including it here for completeness because it reads from all signal data and must be on the read-replica, not the primary database.

3. EXHAUSTIVE SOURCE COUNT

Stream	Named Sources	Count	Status
SIG-NEWS	Google News API, Bank Newsrooms (10), Reuters/Bloomberg, Arab News, Saudi Gazette, Argaam, Maaal	16	SPECIFIED
SIG-REG	SAMA, CMA, Tadawul	3	SPECIFIED
SIG-FIN	Bank IR Pages (10), Fitch, Moody's, S&P;, MAGNiTT, Crunchbase	15	SPECIFIED + OPEN-GAP-SIG-03
SIG-EXEC	LinkedIn Profiles (via Heyreach/Apollo), 11 Named Conferences, Bank Annual Reports	13+	SPECIFIED
SIG-HIRE [NEW]	LinkedIn Jobs, Bayt, Bank Career Pages (10), GulfTalent	13	OPEN-GAP-SIG-01
SIG-VENDOR	7 Named Vendor Websites, Job Posting Tech Stack (cross-link)	8	SPECIFIED
SIG-TENDER [NEW]	Etimad, Manual BD Input	2	OPEN-GAP-SIG-02
SIG-EVENT	11 Named Conferences, LinkedIn Events Tab	12	SPECIFIED
SIG-SUBS	Annual Reports, Subsidiary Websites + LinkedIn (per sub)	varies	SPECIFIED
SIG-PATH	LinkedIn Graph (Decimal Network), Connector Registry	2	SPECIFIED
SIG-PARTNER [NEW]	SIG-NEWS Reclassification, Fintech Saudi Ecosystem	2	OPEN-GAP-SIG-06
SIG-SOCIAL [P2]	Twitter/X API	1	OPEN-GAP-SIG-04 (P2)
INT-ENR	Apollo, Clearbit, Clay, LinkedIn (Heyreach)	4	SPECIFIED
Engagement	Instantly (email), Heyreach (LinkedIn), HubSpot (CRM), Metabase (read-only)	4	SPECIFIED

TOTAL NAMED SOURCES: 95+ individual source endpoints across 15 streams/domains, configured per-account for 10 primary target banks. Every source has: named URL/endpoint, authentication method, per-account query template, cadence, signal type emitted, and EPIS confidence profile.